Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/17/2026
PI Name:	Laurie, Steven Ph.D.		
Project Title:	Investigation of Intracranial Pressure, GLP1-R Agonist, and SANS during Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2024	**End Date:**	12/31/2028
No. of Post Docs:		**No. of PhD Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There are currently no countermeasures proven to prevent the development of spaceflight associated neuro-ocular syndrome (SANS) during long-duration spaceflight or during exposure to the spaceflight analog 6˚ strict head-down tilt bed rest (HDTBR). It has been hypothesized that the unrelenting headward fluid shift and the inability to “stand up” and lower intracranial pressure (ICP) results in a chronic mild elevation in ICP compared to the mean ICP throughout a 24-hour day on Earth. The glucagon-like peptide-1 receptor agonist (GLP-1RA) exenatide demonstrated the ability to lower cerebrospinal fluid secretion in animals and lower intracranial pressure (ICP) in patients with idiopathic intracranial hypertension (IIH) in a phase II clinical trial and has now entered a phase III clinical trial. We propose to confirm the role of elevated ICP in SANS in a strict HDTBR analog by measuring ICP via lumbar puncture and determine whether pharmaceutically lowering ICP would prevent the development of SANS. This task will enroll two groups of 12 healthy subjects to 30 days of strict HDTBR at a bed rest facility chosen by NASA. The intervention group will receive a GLP-1RA (exenatide, 10 µg, twice daily subcutaneous injection) and the control group will receive a placebo (saline, twice daily subcutaneous injection). Approximately two weeks before HDTBR and on the last day of HDTBR, subjects will have a lumbar puncture (LP) to measure ICP. The needle will be attached to a pressure transducer and the signal collected in a data acquisition system. ICP will be measured in resting subjects during standard maneuvers and in multiple body postures to assess changes in ICP and ICP pulse amplitude. Thus, ICP will be recorded continuously for up to several hours to accommodate stabilized measures in head-up, supine, and head-down tilt postures. Intraocular pressure will be measured during ICP measures at each posture to calculate translaminar pressure difference, along with blood pressure to quantify ocular perfusion pressure. Optical coherence tomography (OCT), OCT angiography (OCTA) imaging, electroretinograms (ERG), along with intravenous fluoresceine angiography, will be acquired before and at the end of HDTBR to assess for the development of optic disc edema, optic nerve function, chorioretinal folds, and altered retinal perfusion in both groups. Associations between ICP, OCT outcomes, brain structural changes on MRI, and findings from intravenous fluoresceine angiography will be determined. Group differences will be determined for all outcome measures.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	The impact of SANS on astronaut health is a growing concern. A therapy reducing ICP would be beneficial to alleviate SANS and with the goal of preventing any possible long-term sequelae before it occurs.
Task Progress:	Note: Per the Human Health Countermeasures (HHC) Element, no technical reporting is required for this fiscal year. The necessary international agreements were completed in this annual report period (Ed., 3/17/26).
Bibliography Type:	Description: (Last Updated: 12/10/2025)